

Using Python Packages (Class Activity)

- How to check if a Python package is already available and how to install it using different tools? Pip list or Anaconda to find packages, use pip install and Anaconda to install.
- How can you tell if a package is already available in your Python environment? Pip show or Anaconda
- What happens when you try to use (import) a package that isn't installed?
These will have some ERROR information.
- How do you install a package using:
 - pip? Use pip install ***
 - Anaconda (conda)? Use the GUI to find the package and choose.

Jupyter vs py Files(Class Activity)

- Discuss the difference between Jupyter Notebooks and .py Files .ipynb can show output in page and can use markdown
- What are the advantages and disadvantages of using Jupyter Notebooks?
It can show result immediately, but may not be good at large project.
- When would you prefer using a .py file instead?
In large project and multifile project.
- Which tool do professionals use for large projects? .py
- Form a group and research for 5 minutes and then participate in class discussion

Python's Open-Source Ecosystem (Class Activity)

- Task: (Do it in your group)
- Go to <https://pypi.org> or <https://github.com/trending/python>
- Find one Python package you find fascinating
- Prepare a 1-minute description: pip install time-checker
- What does it do? time-checker is a lightweight and reusable Python package for measuring the execution time of scripts, functions, and code blocks.
- Who might use it? developers who want to profile their Python code quickly
- Why is it interesting It's easy to measure each part of code.

Why Python for Data Analytics? (Class Activity)

- What Makes Python a Great First Tool for Data Analysts? Python is free and easy to build libraries.

- What features make Python accessible to beginners? Easy grammar and show result immediately.
- Why do data professionals keep using it, even when they learn other languages? Convenience to get new libraries and easy to share with team partner.
- How does Python compare to tools like Excel or R? Excel is very easy but difficult for large mount data. R is special and not common use.

Data Ethics & the Responsible Use of Python (Class Activity)

- Scenario: Your team wants to analyze trends by scraping public comments from a social media site like Facebook. The data is public, but users haven't explicitly given permission to be analyzed.
- Discuss the following:
 - Should you proceed? At the beginning, we need to check data is there any private information. Just scrape necessary data.
 - What risks or harms could result from this? Personal information maybe used in bad side. Some time will affect the reputation and social situation.
 - What if the data is used to make business decisions or target users? Risks always here, no one can promise always do the right decision.